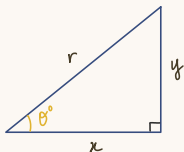


Trigonometry

1 Angles between 0° and 90°



$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

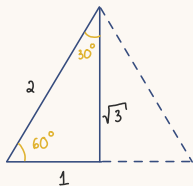
$$\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}$$

$$\tan \theta = \frac{\text{opposite}}{\text{adjacent}}$$

$$\sin \theta = \frac{y}{r}$$

$$\cos \theta = \frac{x}{r}$$

$$\tan \theta = \frac{y}{x}$$



$$\sin 30^\circ = \frac{1}{2}$$

$$\sin 60^\circ = \frac{\sqrt{3}}{2}$$

$$\sin 45^\circ = \frac{1}{\sqrt{2}}$$

$$\cos 30^\circ = \frac{\sqrt{3}}{2}$$

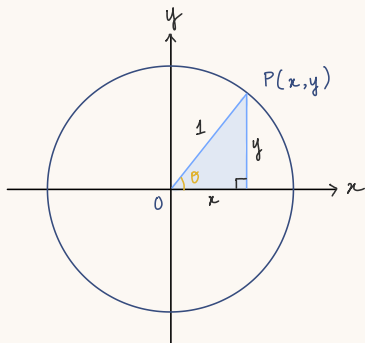
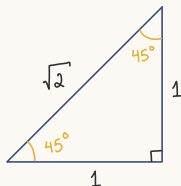
$$\cos 60^\circ = \frac{1}{2}$$

$$\cos 45^\circ = \frac{1}{\sqrt{2}}$$

$$\tan 30^\circ = \frac{1}{\sqrt{3}}$$

$$\tan 60^\circ = \sqrt{3}$$

$$\tan 45^\circ = 1$$



* the unit circle

- center: O (origin)
- radius = 1 unit
- for any **acute angle** between **OP** and **x -axis**:
 a.k.a. basic angle / reference angle

$$\sin \theta = \frac{y}{1} = y$$

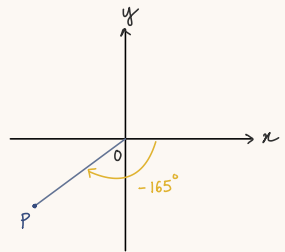
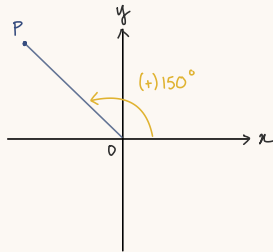
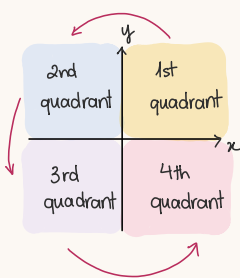
$$\sin 0^\circ = 0$$

$$\cos \theta = \frac{x}{1} = x$$

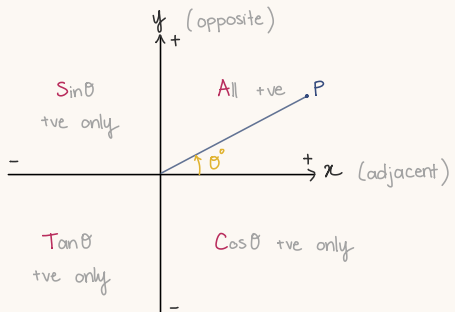
$$\cos 0^\circ = 1$$

$$\tan \theta = \frac{y}{x}$$

2 Angles : calculated anti-clockwise from the x -axis = +ve value



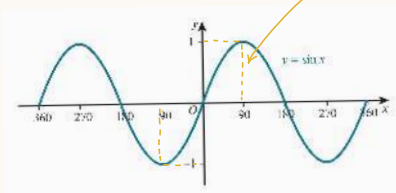
- use a **CAST** diagram to decide whether $\sin$, $\cos$ or $\tan \theta$ is +ve or -ve



3. Trigonometric graphs

$y = \sin \theta$

amplitude : 1



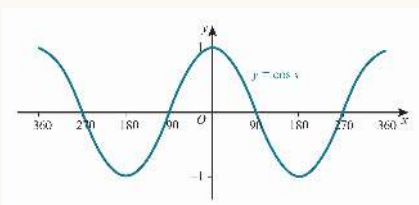
- period: 360° (repeats every 360°)
- rotational order of 2 about the origin (in a 360° , it looks the same 2 times)
- $-1 \leq \sin \theta \leq 1$

minimum

maximum

periodic functions

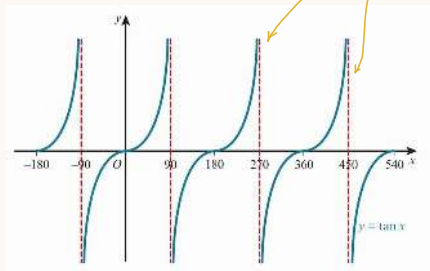
$y = \cos x$



- period: 360°
- symmetrical about y -axis
- $-1 \leq \sin \theta \leq 1$

• $y = \tan \theta$

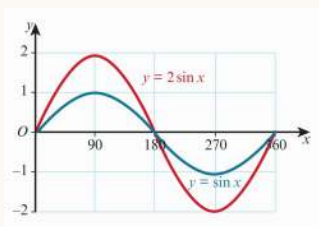
asymptotes: graph gets infinitely close but never touches these lines



- period: 180°
- rotational symmetry 2 about origin
- $\tan \theta$ can be any real number
 - as $\theta \rightarrow 90^\circ$, $\tan \theta \rightarrow \infty$
 - as $\theta \rightarrow -90^\circ$, $\tan \theta \rightarrow -\infty$

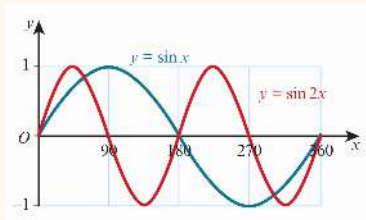
× trig graph transformations

• $y = a \sin x$



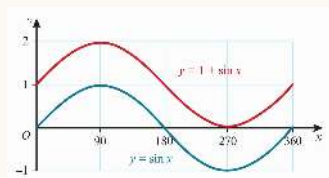
- stretch; factor 2
- // y-axis
- period: 360°

• $y = \sin ax$



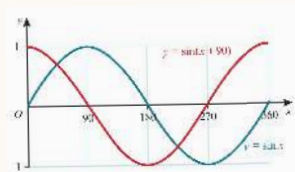
- stretch; factor $\frac{1}{2}$ (= squashed)
- // x-axis
- period: 180°

• $y = a + \sin x$



- translation; of $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$
- amplitude: 1
- period: 360°

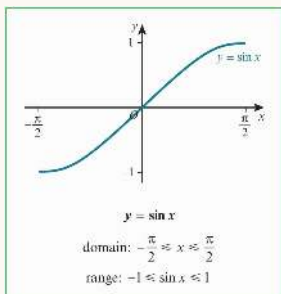
• $y = \sin(x + a)$



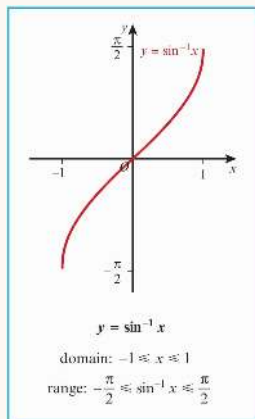
- translation; of $\begin{pmatrix} -90 \\ 0 \end{pmatrix}$
- amplitude: 1
- period: 360°

4. Inverse trig functions

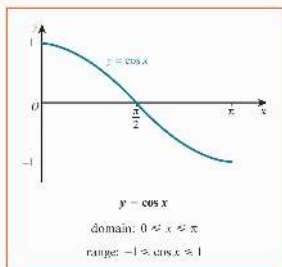
• $y = \sin x$



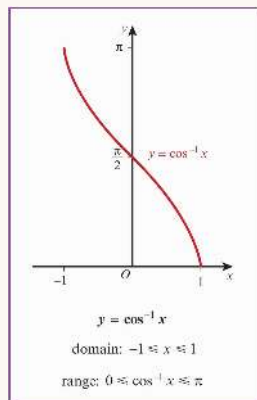
• $y = \sin^{-1} x$



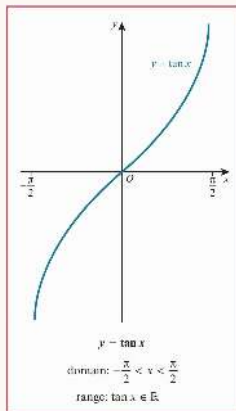
• $y = \cos x$



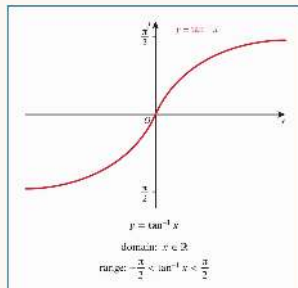
• $y = \cos^{-1} x$



• $y = \tan x$

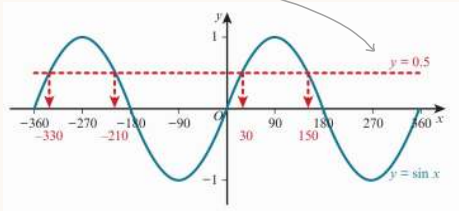


• $y = \tan^{-1} x$



5. Trigonometric equations

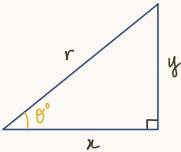
Ex: solve $\sin x = 0.5$ for $-360^\circ \leq x \leq 360^\circ$



• use **calculator** to find $x = 30^\circ$
 ↳ use **symmetry** of graph to find remaining values

↳ 4 values total

* Rule 1.



$$\left. \begin{aligned} \sin \theta &= \frac{y}{r} \\ \cos \theta &= \frac{x}{r} \end{aligned} \right\} \tan \theta = \frac{y}{x} = \frac{\left(\frac{y}{r}\right)}{\left(\frac{x}{r}\right)}$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

for all θ ,
with $\cos \theta \neq 0$

$\equiv$ is an **identity**
 = true for ALL values of x

$$\tan x \equiv \frac{\sin x}{\cos x}$$

* Rule 2

$$x^2 + y^2 = r^2$$

$$\left(\frac{x}{r}\right)^2 + \left(\frac{y}{r}\right)^2 = 1$$

$$\rightarrow \cos^2 \theta + \sin^2 \theta = 1 \quad \text{for all } \theta.$$

$$\sin^2 x + \cos^2 x \equiv 1$$